Muhammad Kaif

AI Engineer • Robotics • Full-Stack Developer

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PROFILE SUMMARY

Artificial Intelligence undergraduate at FAST NUCES building agentic AI systems, computer-vision tools and autonomous robots with Python, C++, Node.js and ROS 2. Shipped LLM agents, voice assistants, n8n automations, an image-retrieval engine with 95% Recall@1 and real-time full-stack backends, and led student teams to deliver automation pipelines and autonomous systems.

EDUCATION

National University of Computer and Emerging Sciences (FAST NUCES) **Expected 2028**
Bachelor of Science in Artificial Intelligence *Islamabad, Pakistan*
Relevant coursework: Data Structures & Algorithms, OOP, Database Systems, Operating Systems, Programming for AI

PROJECTS

SoleMatch: Visual Sneaker Search Engine *(CLIP, ResNet50, FAISS, FastAPI, React, Three.js)* GitHub
- Built an image-to-product retrieval engine comparing CLIP with fine-tuned ResNet50 triplet embeddings indexed in FAISS, reaching **95.0% Recall@1 on 645 held-out queries**; served through a FastAPI retrieval service with a React + Three.js frontend.

Clara: AI Voice Receptionist *(Vapi.ai, n8n, Google Calendar API)* GitHub
- Built an inbound voice agent that answers dental-clinic calls, checks real-time availability in Google Calendar and books appointments through natural conversation, with self-hosted n8n workflows running the scheduling logic.

AI Automation Agents & Lead-Gen Pipelines *(n8n, Groq Llama 3.3, Firecrawl, Tavily)* GitHub
- Designed production-ready agent workflows, including a B2B lead-generation pipeline that scrapes target websites, uses Groq LLMs to analyse business gaps and automatically drafts personalised outreach emails.

Synapse AI Wearable System *(Python, ESP32, Whisper, Qwen, A* Search)* GitHub
- Engineered a wearable Body Area Network with ESP32, GPS and biometric sensors for real-time monitoring; integrated Whisper, Qwen and A* pathfinding to turn sensory data into context-aware AI interactions.

Medidash Autonomous Medical Robot *(ROS 2, Gazebo, MQTT, Robotics Hardware)* GitHub
- Deployed an autonomous healthcare robot integrating custom hardware with ROS 2; configured MQTT machine-to-machine communication to coordinate safe navigation and task execution.

Phantom-Key Touchless Interface *(Python, OpenCV, MediaPipe, Tkinter)* GitHub
- Developed a computer-vision controller that maps hand landmarks to OS commands, with dedicated mouse, keyboard, presentation and media modes; designed the GUI in Figma and Tkinter.

RideFlow Ride-Sharing Platform *(Node.js, Express, MongoDB, SQL, JWT)* GitHub
- Developed a full-stack ride-hailing app with real-time ride matching, JWT authentication, trip lifecycles and a built-in PKR wallet; architected backend workflows with MongoDB, SQL triggers and stored procedures.

Retail Pulse OS *(Python, POSIX Shared Memory, Named Pipes, Multithreading)* GitHub
- Built a multi-process pipeline that ingests and aggregates large retail transaction datasets in real time, using shared memory, named pipes and threads to remove sequential bottlenecks across CPU cores.

Movie Data Manager *(C++, Graphs, Data Structures, OOP)* GitHub
- Modelled 5,044 movies as vertices in a custom graph to power relation mapping, recommendations, search and sorting with optimised time complexity, applying strong object-oriented design.

Interactive 3D Web Experiences *(React Three Fiber, Three.js, GSAP, TypeScript)* Live
- Built a 3D anime-themed chess game with full rules (chess.js), a scroll-driven 3D car showcase with a live paint configurator, and an interactive 3D portfolio with physics-style UI and live GitHub data.

TECHNICAL SKILLS

- **Languages:** Python, C++, JavaScript / TypeScript, SQL, Bash, HTML / CSS
- **AI / ML & Vision:** Scikit-learn, Pandas, NumPy, OpenCV, MediaPipe, CLIP, FAISS, NLP, Whisper
- **Agents & Automation:** LLMs, AI Agents, LangChain, CrewAI, RAG, n8n, Vapi.ai, REST APIs & Webhooks
- **Robotics & Embedded:** ROS 2, Gazebo, ESP32, MQTT
- **Web & Databases:** React, Next.js, Node.js, Express, FastAPI, Three.js, MongoDB, PostgreSQL, MySQL, Pinecone
- **Tools:** Git & GitHub, Linux (Ubuntu / WSL), Docker, VS Code, Visual Studio, Postman

ACTIVITIES & LANGUAGES

- **Programming for AI Project**, FAST NUCES | **Member**, Welfare of Humanity (WOH) Society, FAST NUCES
- **Languages:** English (Fluent), Urdu (Native), German (Beginner)